PureGym – Applying NLP for topic modelling in a real-life context

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Date: 5th July 2025

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INTRODUCTION

The goal of this project was to analyse customer reviews from Google and Trustpilot for a leading gym company, with the aim of extracting actionable insights to guide business improvements. By leveraging both traditional and modern natural language processing (NLP) techniques—including Gensim LDA, BERTopic, and large language models (LLMs) from Hugging Face—the analysis sought to systematically identify, cluster, and interpret the principal themes of customer dissatisfaction.

PROBLEM STATEMENT

The gym company has received a large volume of customer reviews across platforms such as Google and Trustpilot. A significant portion of these reviews express dissatisfaction with various aspects of the service. However, these negative reviews are **unstructured** and lack categorization, making it difficult for the business to understand and act on the key issues impacting customer experience.

The goal of this project is to **analyse negative customer reviews** using Natural Language Processing (NLP) techniques to identify the **main causes of dissatisfaction**, assess **emotional intensity**, and extract **actionable insights**. By leveraging advanced models like **BERTopic**, **LDA**, and **emotion classification with BERT**, as well as **LLM-powered topic summarization**, this analysis seeks to:

- Detect the **most common issues** customers complain about (e.g., billing, cleanliness, staff behaviour).
- Understand the **emotions** driving negative feedback, particularly anger and frustration.
- Prioritize key problem areas by analysing emotion-tagged feedback.
- Provide **data-driven recommendations** for operational and service improvements.

This will help the gym company enhance customer satisfaction, reduce churn, and improve its overall brand perception.

OVERVIEW APPROACH

We aimed to analyse negative customer reviews from Google and Trustpilot to uncover actionable insights for a gym company. The overarching objective was to construct a robust natural language processing (NLP) pipeline for text data, train machine learning models for sentiment classification, and gain deeper insights into the underlying themes within review data using topic modelling.

The analysis pipeline combined traditional

1. NLP techniques, emotion classification,
2. Topic modelling (BERTopic, LDA), and
3. Large language models (**microsoft/Phi-4-mini-instruct instead of** Falcon-7B due to very high computation requirements)

These were used to identify recurring issues and emotional triggers among dissatisfied customers.

Key stages included:

1. Data preprocessing,
2. Sentiment filtering,
3. Emotion detection,
4. Semantic topic extraction using BERTopic and Falcon. Emotion-based filtering (specifically anger) helped isolate the most critical complaints and
5. LDA topic modelling

DESCRIPTION OF THE ANALYSIS

Data Source: Reviews from Google and Trustpilot

Preprocessing:

- Lowercasing, stopword removal, lemmatization, Punctuation and Digit Removal
- Tokenization for LDA

Models Used:

- **BERTopic**: Captures semantic clusters using transformer embeddings
- **Phi-4-mini-instruct (Instead of Falcon-7B)**: Extracts top 3 topics from each review using prompt-based generation
- **Gensim LDA**: Provides interpretable word-based topics
- **Emotion Classifier**: bhadresh-savani/bert-base-uncased-emotion for emotion tagging

Emotion Aggregation:

- Each review was labelled with the top predicted emotion
- Angry reviews were extracted for focused topic modelling

Visualizations:

- Emotion distributions (bar plots)
- BERTopic visualizations (distance map, bar chart)
- LDA pyLDAvis (for inter-topic distances and salient terms)

EXPLANATION OF VISUALISATIONS

1. Frequency Distribution and Word Cloud

- **Purpose**: To Show how **often each topic or word appears** in your dataset and provides a **visual summary** of the most frequent words or topics
- **How**: Flatten your list of topics or tokens, makes it easy to **spot repeated terms** in a creative, non-tabular way.
- **Output**: Visualize the counts using a **bar chart or histogram and** A cloud where **larger words = higher frequency**.

2. BERTopic Distance Map

- **Purpose**: To understand the **semantic relationship** between generated topics.
- **How**: BERTopic uses dimensionality reduction (e.g., UMAP) to project topic vectors into 2D space.
- **Output**: A scatter plot where each point represents a topic, and proximity indicates similarity.

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2. BERTopic Bar Charts (Top Terms per Topic)

- **Purpose**: To display the most representative terms for each identified topic.
- **How**: For each topic, BERTopic ranks words by relevance and frequency.
- **Output**: Horizontal bar charts showing top words like refund, cancel, machine, dirty, membership, etc.

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3. BERTopic Heatmaps

- **Purpose**: To visualize and understand the relationships between topics.
- **How**: For each topic, BERTopic ranks words by relevance and frequency.
- **Output**: Shows **which words define each topic,** Helps interpret the **meaning and context** of topics.

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4. Emotion Distribution Bar Charts

- **Purpose**: To visualize the primary emotions, present in negative reviews from both Google and Trustpilot.
- **How**: Each review was passed through a BERT-based emotion classification model (bhadresh-savani/bert-base-uncased-emotion). The top-scoring emotion for each review was captured.
- **Output**: A bar chart showing frequency counts for emotions like anger, sadness, fear, joy, etc.

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5. Topic Frequency Histogram (LLM Topics)

- **Purpose**: After extracting top 3 topics per review using the Phi-4-mini LLM, the full list was flattened and visualized.
- **How**: Counted frequency of each topic across all reviews.
- **Output**: A histogram showing which LLM-derived topics appeared most frequently.

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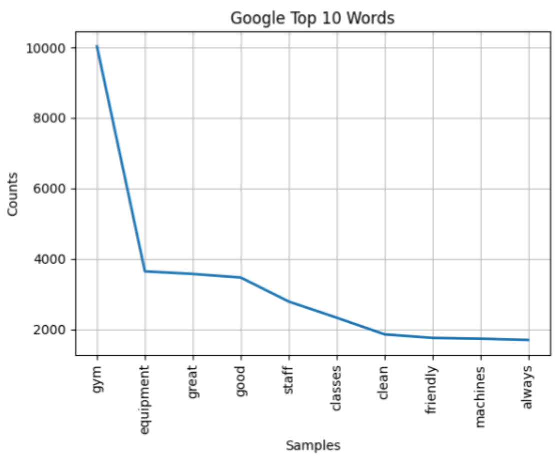
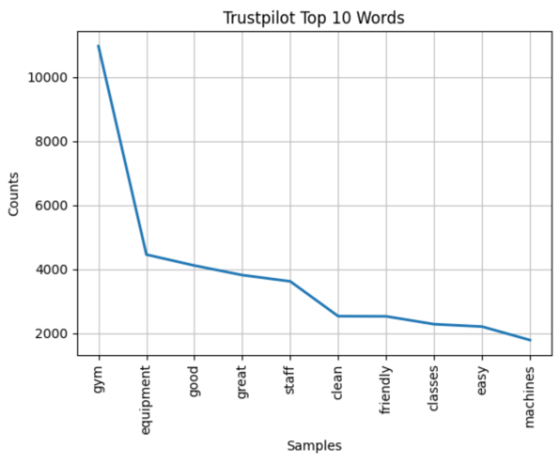
6. LDA pyLDAvis Visualization

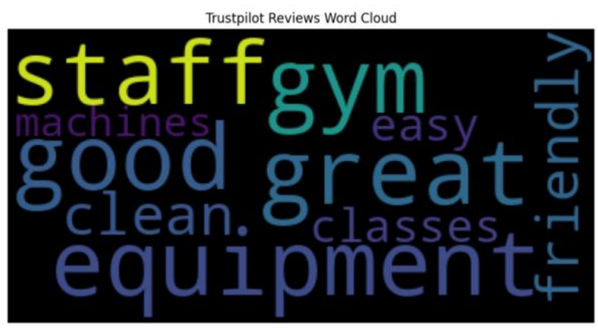
- **Purpose**: To compare Gensim's Latent Dirichlet Allocation (LDA) topics with BERTopic results.
- **How**: Used pyLDAvis to show:
    - **Topic circles**: Representing each topic and its size (prevalence).
    - **Salient keywords**: For each topic on hover or in sidebar.
- **Output**: Interactive plot showing inter-topic distances and keyword contributions.

DATA VISUALISATION

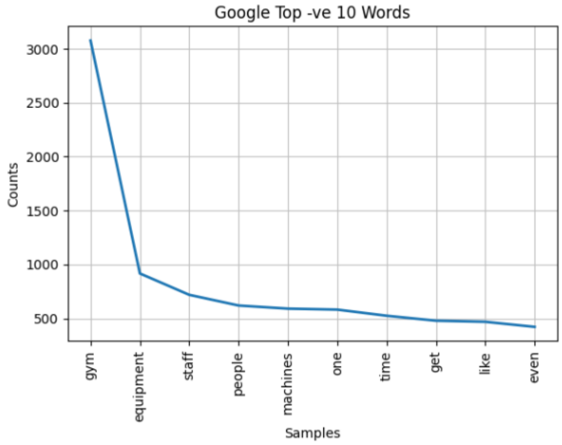
1. Top 10 Reviews

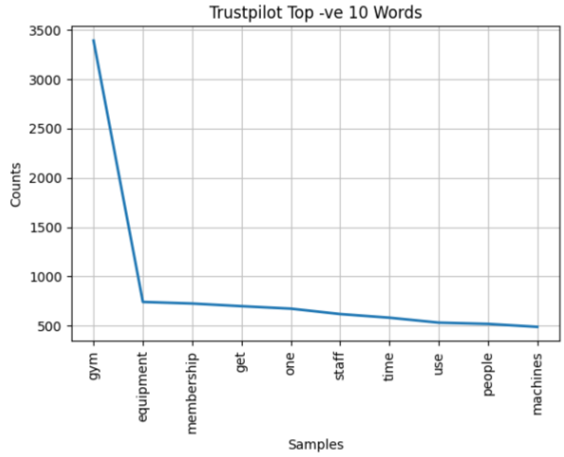






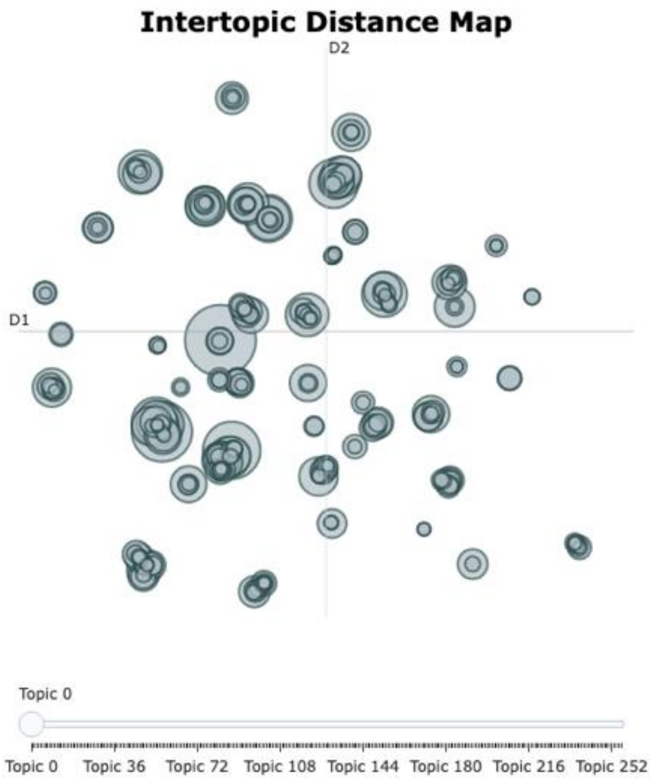
2. Top 10 Negative Reviews





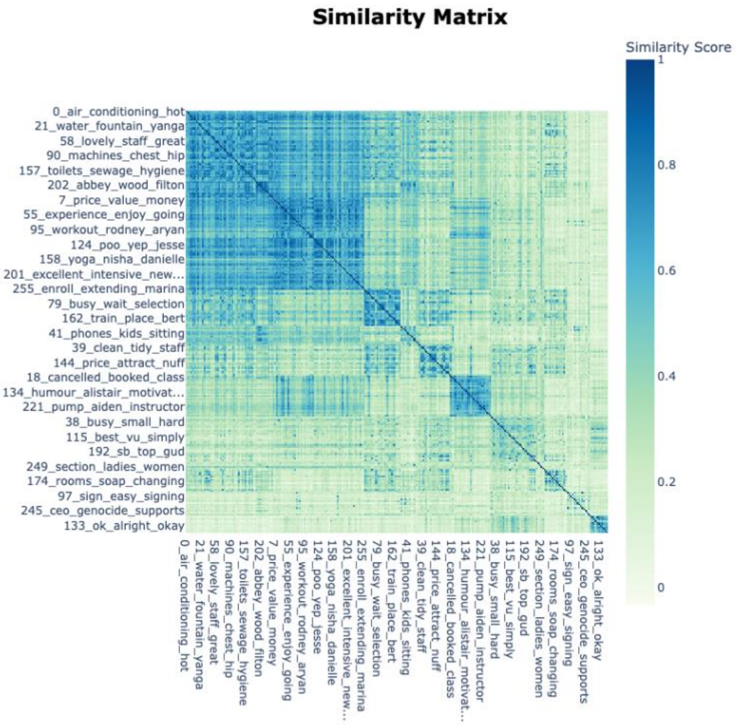
3. BERTopic all reviews

- **BERTopic Distance Map**

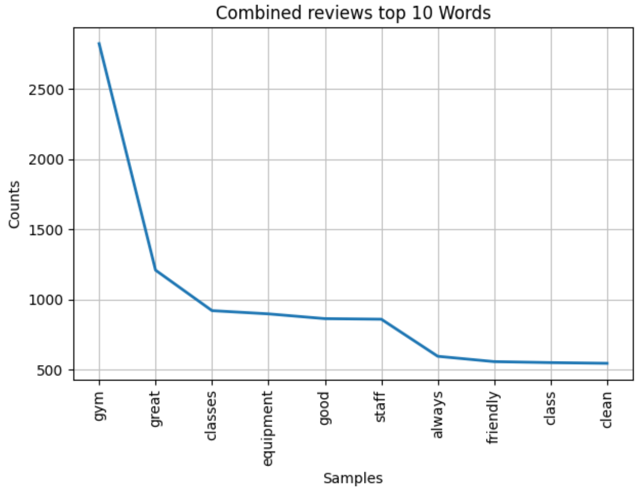


- **BERTopic Bar Charts**

- **BERTopic Heatmaps**

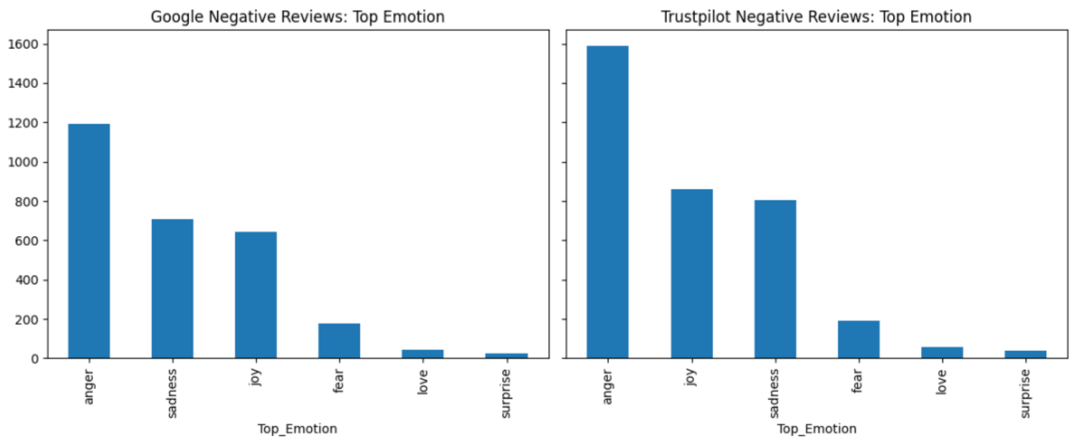


4. Combined reviews for top 30 locations



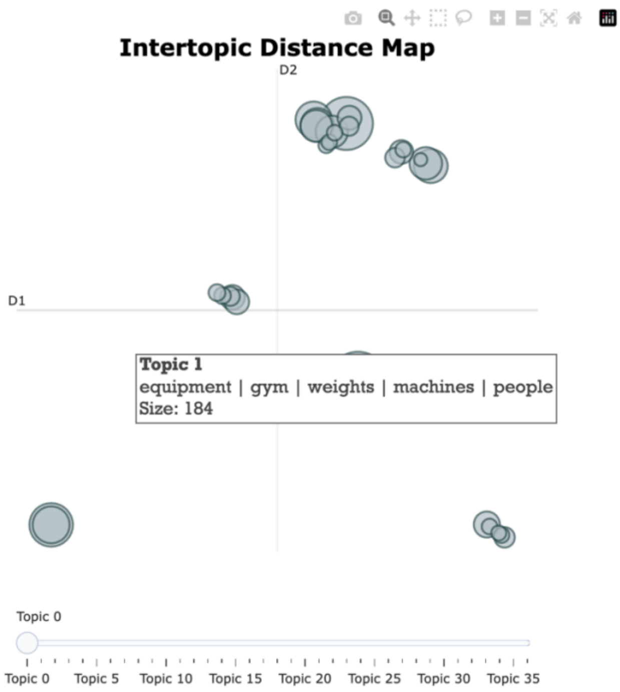
5. Emotion Analysis for Top Emotion - Anger

- **Emotion Distribution Bar Charts**

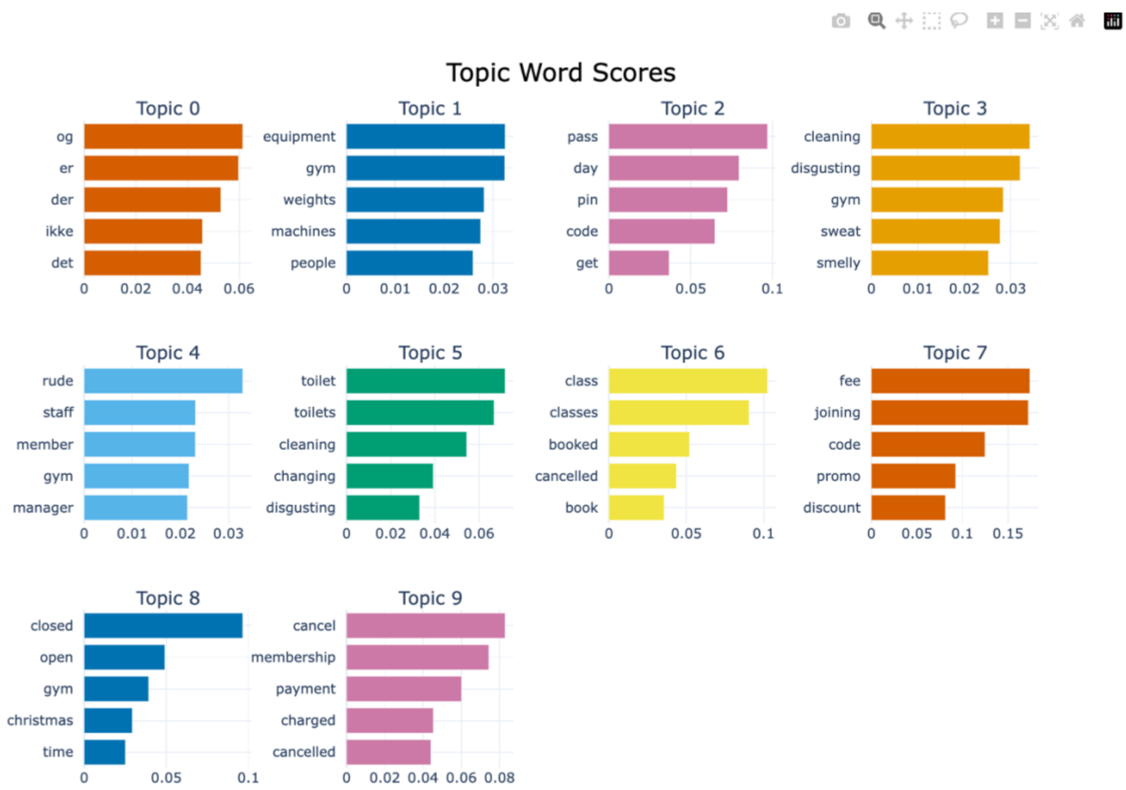


6. BERTopic Anger emotion

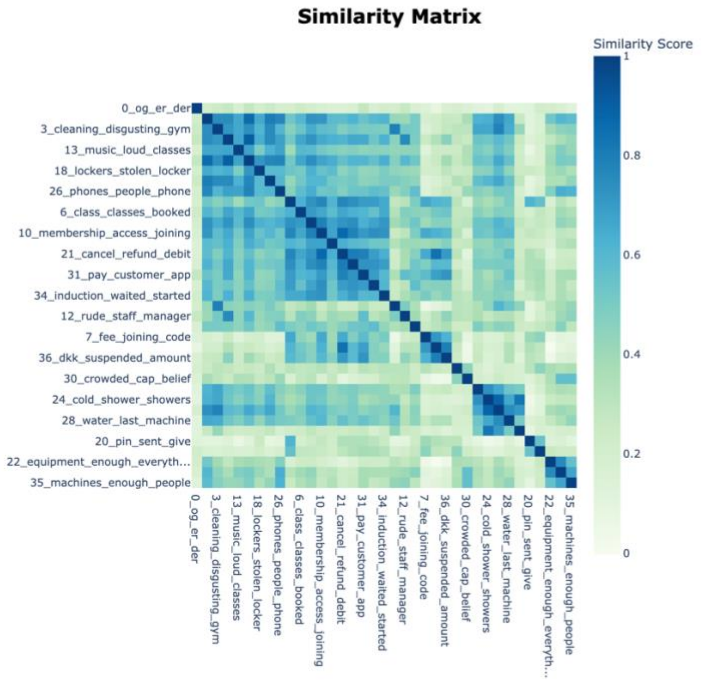
- **BERTopic Distance Map**



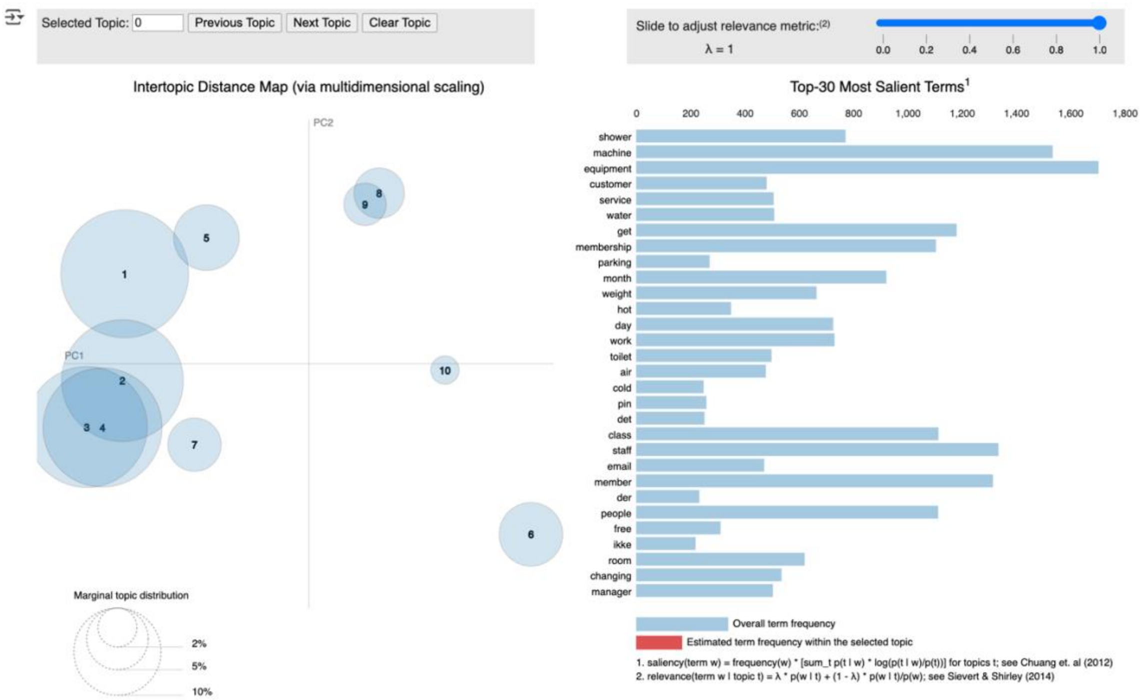
- **BERTopic Bar Charts**



- **BERTopic Heatmaps**



7. Gensim LDA (LDA pyLDAvis Visualization)



CONCLUSION

The analysis of negative reviews using BERTopic, LDA, and Phi-4-mini revealed consistent themes: **rude staff**, **broken equipment**, **unclean facilities**, **membership issues**. BERTopic and LLM-based extraction provided clearer, more actionable groupings. Emotion analysis showed **anger** as the dominant negative emotion, especially in reviews about staff behaviour and cancellation difficulties.

BERTopic produced clearly defined and interpretable clusters, while LDA identified similar but broader themes. Visualizations such as frequency histograms and word clouds helped quantify and illustrate the most common pain points, with “rude staff” and “broken equipment” being the most frequent. Heatmaps and UMAP plots showed tight clusters around facility and service issues. Overall, the combination of models and visual tools confirmed that operational weaknesses—especially around staff behaviour, maintenance, and communication—are driving customer dissatisfaction.

INFERENCES

- **Thematic Consistency**: Across all modelling approaches, the main pain points for customers were consistent—cleanliness, staff attitude, equipment issues, membership/billing, and overcrowding.
- **Model Complementarity**: LLMs provided focused, human-readable topics; LDA offered probabilistic word distributions and surfaced some sub-themes; BERTopic delivered clear, actionable clusters when fed with LLM-extracted topics.
- **Business Relevance**: The actionable insights generated are directly tied to the most frequent and critical customer concerns, offering a clear roadmap for targeted improvements.
- **Visualization Value**: Topic distance maps and salient term bar charts aided in communicating findings to stakeholders and prioritizing interventions.